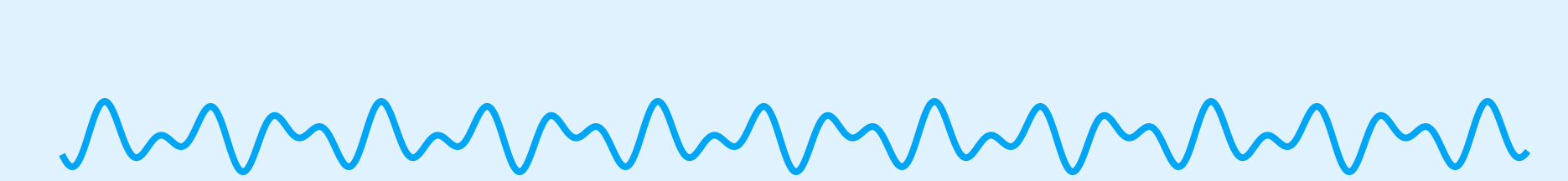
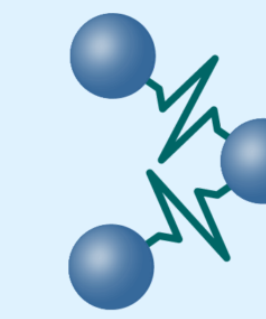


GATHERING INSIGHTS INTO CFDE PROJECTS

Integration and Coordination Center (ICC)



MOTIVATION

The **ICC** is tasked with (among other things) gathering and presenting information about CFDE projects. We want to allow project **groups** and the **NIH** to easily see the activity and impact of their work in a simple and centralized way.

The info we gather currently includes publications, software stats, and website metrics. These come from a variety of sources, take many forms and sizes, and can be sensitive or private, which presents several challenges that led us to develop a unique and flexible software architecture to handle them.

Code discussed here is viewable in the public GitHub repo:

CHALLENGES

Frequency	Need to gather info ideally daily
Availability	APIs available for some info, but some need webpage scraping
Automation	Some (very valuable) info is not available programmatically at all
Scale	Large # of projects, queries can take a long time
Complexity	Info is complex, dense, and different structures and formats
Time Burden	Groups don't have time to set up complex integration
Transparency	Everyone should be able to see <i>how</i> all info is gathered
Access	Some info should be restricted to certain people/groups

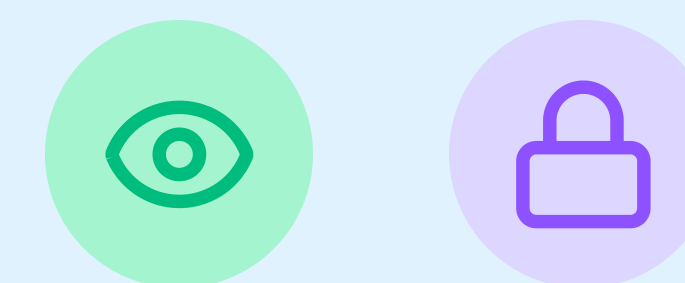
TECHNOLOGIES



TypeScript	To maintain full type safety from gathering to displaying
Node	To run gathering process and other parts of pipeline
Vue	To create live web dashboard (better than re-emailing PDFs)
GitHub Actions	To automatically run pipeline, tests, deployment, etc.
Playwright	To scrape HTML pages (and print dashboard as PDFs as needed)
Various libraries	To parse CSV, PDF, etc.
Custom tools	To auto-parallelize and cache (w/ TTL) all network requests
API clients	To ease gathering from e.g. GitHub and Google Analytics
Django	To create backend database and API to serve the dashboard

STRATEGY

To address *transparency*, we devised a **public** & **private** repo split. **Public** has (virtually) all of our code and the non-sensitive info. **Private** is barebones: it has one basic GitHub Actions workflow (to copy and run the exact code in **public**) and the sensitive info.



To address *time burden*, we've devised ways to keep the manual steps groups need to perform *as minimal as possible*:

Google Analytics	"Invite" our account to your properties Add custom "tags" to properties w/ your core project #s
GitHub	Tag your repos w/ core project #s

PIPELINE

nih-cfde/icc-eval-core public repo	nih-cfde/icc-eval-core-private private repo, requires invite
Start on daily schedule	Check out (copy) public repo locally
Run gather process in <i>public</i> mode	Get GH Actions secrets (API keys)
	Run gather process in <i>private</i> mode
Funding Opportunities	Google Analytics
ID, activity code, etc. commonfund.nih.gov	Get all sites we have access to
Projects	Match sites to projects by tag
#, name, award \$, dates, etc. reporter.nih.gov	Gather # of users/sessions over time, top countries, etc.
Publications	GitHub
Title, date, citations, etc. iCite, RePORTER	Search GitHub for any repo tagged with CFDE project #
Journals	Match repos to projects by tag
Name, rank, ISSN, etc. E-utilities, Scimago	Gather stars, PRs, commits over time, languages, etc.
Store [public] results in repo	Store [private] results in repo
Raw & normalized/pruned JSON	Raw & normalized/pruned JSON
Trigger private repo	Build dashboard and deploy to live site
	(just functionality & public info, no private)

ACCESS

To ensure appropriate access to private info, we have implemented a standard architecture with social sign-on authentication:

Architecture	User's POV
Private Repo JSON	Go to live, public dashboard
↓	Browse public info
Backend Database	Try to look at your project
↕	FE makes request to BE for private info
Backend REST API	Access denied, request fails
↔	Click "log in"
Frontend Dashboard	FE makes request to BE
↕	Redirected to ORCID, log in
ORCID Auth	BE returns auth cookie
	Browser saves cookie, FE can't see it
	Try to access your project again
	Browser auto-attaches auth cookie
	BE checks if your ORCID has access
	BE returns (private) project info
	FE displays it in dashboard

RESULTS

Our method is fully automatic, runs quickly and reliably, and is flexible enough to gather many other types of info in the future. The dashboard has received positive feedback from project groups and NIH staff, successful in providing a convenient and effective way to see project impact across the CFDE.

